MILITARY SPECIFICATION

CLOTH, ACETATE, FLUORESCENT

1. SCOPE

1.1 Scope.- This specification establishes the requirements for fluorescent dyed acetate cloth used in the manufacture of flags for dan buoys and master reference buoys.

1.2 Colors.- The fluorescent acetate cloth shall be furnished in the following colors as specified (see 6.1):

Color R - Daylight Fluorescent Red.
Color Y - Daylight Fluorescent Yellow.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
CCC-T-191 - Textile Test Methods.
PPP-P-1133 - Packaging and Packing of Synthetic Fiber and Silk Fabrics.

STANDARD

FEDERAL
FED-STD-4 - Glossary of Fabric Imperfections - Woven Synthetic Fabrics.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Hardy - Handbook of Colorimetry.

(Published by the Technology Press, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

TEXTILE FIBERS PRODUCTS IDENTIFICATION ACT

(Application for copies should be addressed to the Federal Trade Commission, Washington, D.C. 25380.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction Sample.- Unless otherwise specified in the contract or order, (see 6.1) a preproduction sample of the fluorescent cloth shall be manufactured using the facilities and procedures proposed

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:23Z
Check the source to verify that this is the current version before use.

for production. The sample shall be tested in accordance with Section 4 for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2 Material.- The material used in the manufacture of fluorescent acetate cloth shall be bright non-pigmented acetate, fluorescent dyed so that the finished product shall meet the requirements of this specification.

3.3 Weave.- The weave shall be plain with single picks in the filling and two warp ends weaving as one.

3.4 Properties.- The properties of the finished cloth shall be as specified in table I and in the following paragraphs when tested by the indicated method of CCC-T-191 (see 4.5.2).

Table I - Properties of Fluorescent Acetate Cloth

Characteristic	Requirement
Weight, ozs. per sq yd	7.3 ± 0.3
Yarns per inch, min	
Warp	100
Filling	38
Breaking strength ravel strip method, lbs, min	
Warp	78
Filling	70
Water resistance, spray rating, min	90, 90, 80
Colorfastness to:	
Crocking	Pass
Accelerated Weathering	Good
Salt Water	Good

3.5 Dimensions.-

3.5.1 Width.- Unless otherwise specified (see 6.1), the minimum width of the cloth including commercial width selvages, shall be 39 inches.

3.5.2 Length and put up.- Unless otherwise specified (see 6.1), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up on full width rolls and marked as specified in PPP-P-1133.

3.6 Color.- The cloth shall be piece dyed; it shall be furnished in one of the following listed colors as specified by the procuring activity (see 6.1) and the appearance to the eye in daylight or visible white light shall be as listed alongside the color name.

Daylight Fluorescent Red - purplish red
Daylight Fluorescent Yellow - yellowish orange

3.6.1 Spectrophotometric properties.- The color of the daylight fluorescent cloth shall conform to either the spectrophotometric properties or to the alternate spectrophotometric properties specified in Table II, as applicable for the color specified by the procuring activity (see 6.1) when determined as specified in 4.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:23Z
Check the source to verify that this is the current version before use.

Table II - Spectrophotometric Properties of Daylight Fluorescent Cloth Colors

Wavelength of light reflected and emitted, millimicrons	Reflectance expressed as a percentage of light which would reflect from magnesium oxide			
	Red		Yellow	
	min	max	min	max
400	12	25		14
420				12
440	16	30		
450	20	35		
480			25	35
490			22	35
520			12	24
530		8	10	22
540		8		
550		8		
560				
580			115	
590			120	
600			115	
610				
620			90	
630	135			
640	135		75	
650	135			
680	105		65	

Alternate Spectrophotometric Properties of Daylight Fluorescent Cloth Colors		
Characteristic of light reflected and emitted	Red	Yellow
Dominant wavelength, millimicrons		580 to 589
Complement of dominant wavelength, millimicrons	490 to 500	
Purity, percent	57 to 66	65 to 75
Brightness, percent, minimum	23	60

3.7 Workmanship. - Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material certification. - Certificates showing conformance with the applicable material specification shall be available to the inspector for each lot of cloth. Certification shall contain actual test, examination, or other verifiable quality data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:23Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections.- Inspections of the fluorescent acetate cloth shall be classified as follows:

(a) Preproduction inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Sampling.-

4.3.1 Preproduction sample.- Prior to beginning production a sample 4 yards in length, full width, manufactured in accordance with 3.1 shall be submitted or made ready for the contracting officer or his authorized representative for inspection in accordance with 4.4.

4.3.2 Quality conformance inspection sampling.-

4.3.2.1 Lot.- For purposes of sampling for examination and tests, a lot shall consist of all cloth made under essentially the same conditions and offered for delivery at one time.

4.3.2.2 Sampling for visual and dimensional examination.- A random sample of cloth shall be selected from each lot for visual and dimensional examination in accordance with table III. No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls.

Table III - Sampling for visual and dimensional examination

Lot size in yards	Sample size in rolls 1/
Up to 1300	3
1301 to 3200	5
3201 to 800	7
8001 to 22000	10
22,001 to 110,000	15
110,001 and over	25

1/ If a lot contains fewer than 3 rolls, each roll in the lot shall be examined.

4.3.2.3 Sampling for tests.- Samples for testing shall be selected in accordance with table IV. The unit of product for testing shall be 3 yards, full width, of the finished cloth. Random sampling shall be accomplished so that, if possible, each 3 yard sample shall be taken from a different roll and suitably tagged to identify it with the roll sampled.

Table IV - Sampling for tests

Lot size in yards	Number of samples	Max. number of defects available in sample
Up to 800	2	0
801 to 22,000	3	0
22,001 and over	5	1

4.4 Preproduction inspection.-

4.4.1 Preproduction examination.- The preproduction sample selected in accordance with 4.3.1 shall be subjected to visual examination for appearance, color and finish as specified in 4.5.1.

4.4.2 Preproduction testing.- The preproduction sample selected in accordance with 4.3.1 shall be subjected to the tests specified in 4.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:23Z
Check the source to verify that this is the current version before use.

4.5 Quality Conformance Inspection.-

4.5.1 Visual and dimensional examination.-

4.5.1.1 Yard by yard examination.- Thirty yards from each roll in the sample selected in accordance with Table III shall be examined on the face side. All defects as defined in Section III of FED-STD-4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed below. No linear yard (increment of one yard on the measuring device of the inspection machine) shall be penalized more than 4 points. The lot shall be unacceptable if the points per 100 square yards exceed 25.0. Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size x 3600}}{\text{Contracted width of cloth (in) X sample size (yds)}} = \text{Points per 100 sq yds}$$

Demerit points - Demerit points shall be assigned as follows:

For defects 3 inches or less in any direction - one point
For defects exceeding 3 inches, but not exceeding 6 inches in any direction - two points
For defects exceeding 6 inches, but not exceeding 9 inches in any direction - three points
For defects exceeding 9 inches in any direction - four points

The following defects when present shall be scored 4 points for each yard in which they occur:

Baggy, ridgy, or wavy cloth
Objectionable odor
Width of cloth less than minimum specified
Poor dye penetration, mottled, streaky or cloudy

4.5.1.2 Examination for identification of pre-shrinkage process and compliance with Textile Fiber Products Identification Act.- During the yard by yard examination, each roll in the sample shall be examined. The lot shall be unacceptable if two or more rolls in the sample contain identification of the preshrinkage process by name or trademark on the cloth or the ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.5.1.3 Examination for length.- During the yard by yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length. The lot shall be unacceptable if the total of the actual length of rolls in the sample is less than the total of lengths marked on the ticket.

4.5.2 Test procedures.- Each sample of cloth selected in accordance with table IV shall be tested with the methods indicated in table V.

Table V - Methods of testing

Characteristic	Requirement Paragraph	Test Method CCC-T-191
Weight, oz per sq yd	Table I	5041
Yarns per inch	Table I	5050
Breaking strength	Table I	5104
Water resistance, spray	Table I	5526
Colorfastness	Table I	5651, 5670, 5632
Weave	3.3	Visual
Spectrophotometric properties	3.6.1 & Table I	Par. 4.5.2.1 or 4.5.2.2 (herein)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:23Z
Check the source to verify that this is the current version before use.

4.5.2.1 Spectrophotometric properties.- Specimens of cloth shall be cut out at a distance of at least 6 inches in from the selvage and each end of the sample. The specimens shall be cut from samples which have been protected from exposure to light. A color analyzer shall be used. The specimen shall be directly illuminated, normal to the surface exposed, by the full spectral emission from a light source having the spectral quality of CIE (Commission Internationale de l'Eclairage or International Commission on Illumination) Source "C". The light reflected and emitted by the fluorescent material shall be picked up with a cylindrical mirror type apparatus as shown in Figure 1 and shall be evaluated spectrophotometrically with the instrument using a slit width of 0.1 mm. The Beckman DU spectrophotometer, the Beckman DK spectrophotometer, or or equivalent may be used as the instrument for obtaining the spectral data. The light shall be measured at wave lengths specified in Table II and expressed in terms of a percentage of the light which would be reflected from the reference standard, a freshly prepared magnesium oxide surface, under similar conditions. To minimize the effects of directional reflectance, readings at two (2) positions shall be averaged (the original sample position and a position where the sample has been rotated through 90 degrees). Readings shall be taken on the face side of the cloth specimen backed-up by three (3) thicknesses of the cloth sample to minimize differences in cloth texture and background.

4.5.2.2 Alternate spectrophotometric properties. As in 4.5.2.1 the percent reflectance shall be determined for the fluorescent cloth by measuring light reflected and emitted at each of the following 30 wavelengths: 422, 432, 436, 439, 444, 450, 456, 461, 462, 469, 478, 489, 495, 515, 530, 541, 544, 552, 562, 564, 573, 577, 585, 589, 600, 601, 611, 624, 627, and 646 millimicrons. The measurements at 436, 461, 544, 564, 577, 589, 600, 611, 624, and 646 millimicrons shall be added together. The sum shall be miltiplied by 0.098. The product shall equal "X".

The measurements at 489, 515, 530, 541, 552, 562, 573, 585, 601, and 627 millimicrons shall be added together. The sum shall be multiplied by 0.100. The product shall equal the percent brightness "Y" of the reflected and emitted light.

The measurements at 422, 432, 439, 444, 450, 456, 462, 469, 478, and 495 millimicrons shall be added together. The sum shall be multiplied by 0.118. The product shall equal "Z".

Dominant wavelength and purity of the light reflected and emitted by the daylight fluorescent dyed cloth shall be determined by consulting charts in the Handbook of Colorimetry by Hardy, after values for x and y are calculated with the following equations:

$$x = \frac{X}{X + Y + Z} \qquad y = \frac{X}{X + Y + Z}$$

In the case of the daylight fluorescent red color, the complement of the dominant wavelength is numerically identifical with the dominant wavelength determined by the above method.

4.6 Examination for preparation for delivery requirements.- An examination shall be made to determine that packaging, packing, and marking requirements of Section 5 of this specification are complied with.

5. PREPARATION FOR DELIVERY

(Preparation for delivery requirements specified herein apply only to direct Government procurment. Preparation for delivery requirements between contractors and sub-contractors shall be as specified in the individual order.)

5.1 Domestic shipment and early use.-

5.1.1 Preservation and packaging.- Preservation and packaging, which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity.

5.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier, at lowest rate, and will afford protection against damage during shipment from the supply source to the using activity. The shipping container or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:23Z
Check the source to verify that this is the current version before use.

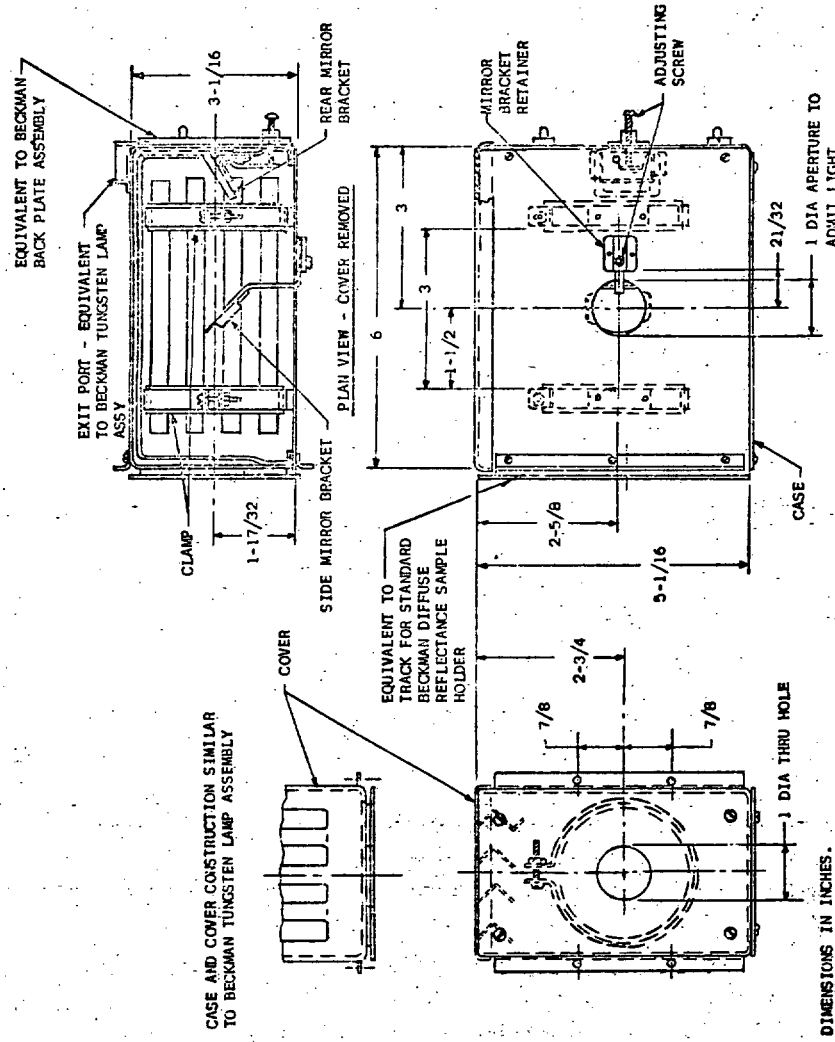


Figure 1 - Cylindrical mirror apparatus

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:23Z
Check the source to verify that this is the current version before use.

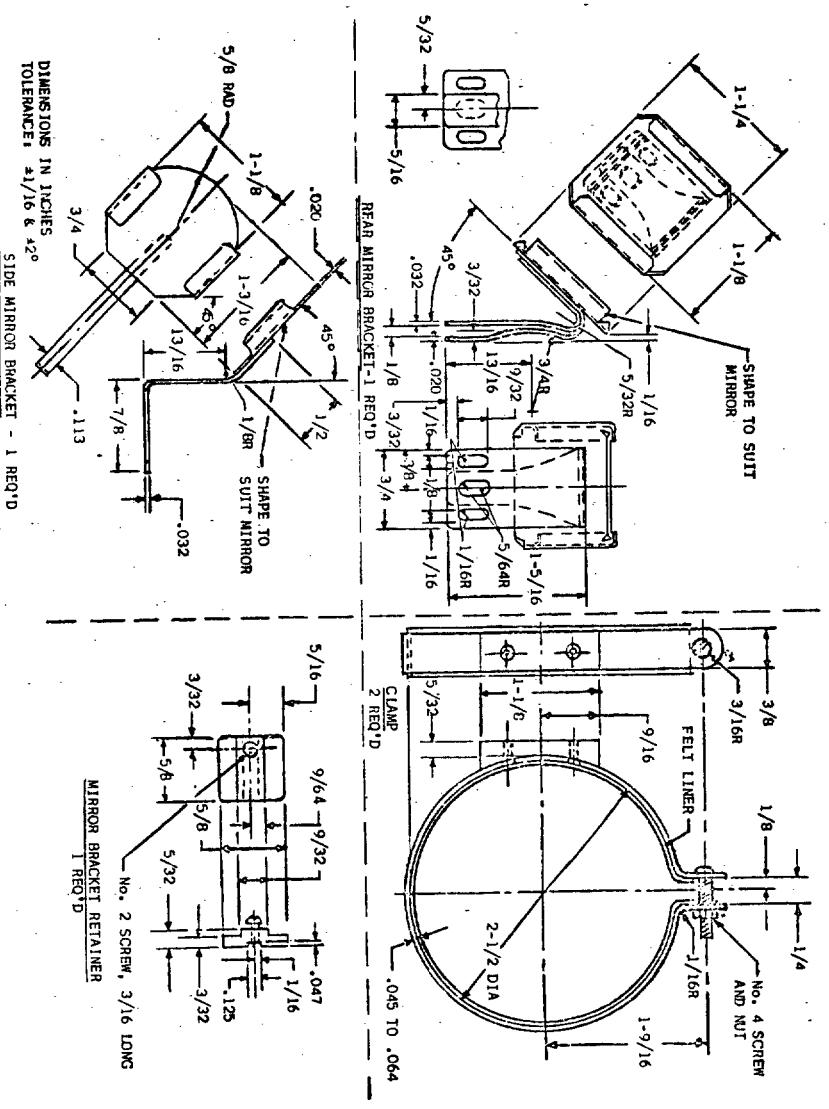


Figure 1 - Cylindrical mirror apparatus (continued)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:23Z
Check the source to verify that this is the current version before use.

5.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and delivery destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.1).

5.2.1.1 Preservation and packing. -

5.2.1.1.1 Levels A and C. - The cloth, put up as specified, shall be packaged in accordance with the applicable requirements for rayon banner cloth of PPP-P-1133.

5.2.1.2 Packing. -

5.2.1.2.1 Levels A, B, and C. - The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.2.1.3 Marking. - In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1133.

6. NOTES

6.1 Ordering Data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color (see 1.2, 3.6, and 3.6.1).
(c) Whether preproduction sample is required (see 3.1).
(d) Dimensions and roll put-up if other than as specified (see 3.5).
(e) Preservation, packaging, packing, or marking requirements other than those required by paragraph 5.1 (see 5.2).

6.2 Preproduction. -

6.2.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - SH
(Project 8305-N016)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:23Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

(NAVSHIPS 0VPRT 12-66)

819500

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:23Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL SHIP ENGINEERING CENTER
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER, NAVAL SHIP ENGINEERING CENTER
FEDERAL & MILITARY DOCUMENTS & DOD STANDARDIZATION BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:23Z
Check the source to verify that this is the current version before use.